2.4 Stage 2: voxel-wise Bayesian tumor segmentation

From localization to segmentation. The two stages answer different questions. Stage 1 is *localization*. A tumor agent can tell that a region looks abnormal but not where it lies anatomically, nor whether that location makes sense for the cancer; this needs the anatomy agents and the knowledge base. Stage 1 therefore reasons globally, over relations between tumor units and organs, and outputs for every lesion L a coarse decision (accepted, uncertain or rejected), its anchors with their candidate extents and extent priors, and a host credible set $\mathcal{S}_L$ with probabilities $P(h \mid L)$. Stage 2 is *segmentation*. It works locally, voxel by voxel, on every accepted and uncertain lesion: where exactly the tumor is, which voxels the agents included wrongly or missed, and how the tumor is composed. The agents recognise tumor as a whole; they do not delineate its internal structure.

Stage 2 is agent-agnostic. It reads only the outputs of stage 1, never the agents' masks directly, so adding an agent changes stage 1 (its claims, anchors and supports) but not the model below. Between the stages the role of the host changes. In stage 1 the host is unknown and inferred (Eq. (5)); in stage 2 it is *known with uncertainty*: $P(h \mid L)$, $h \in \mathcal{S}_L$, renormalised over the credible set, is a fixed prior, and the anatomy stays cut. Stage 1 does not fix the tumor at voxel level. Its extents are proposals drawn by the agents, its decision concerns the lesion as a unit, and a single-agent anchor may be wrong. The tumor mask is therefore a latent variable of stage 2, with a prior learned from expert data, and no voxel is fixed to tumor in advance.

Stage-1 extent marginal. For an anchor a with candidate extents $\mathcal{X}_a$ and extent prior w_a^0 (Eq. (9)), the probability under that prior that voxel v belongs to the tumor drawn around a is

$$q_v^a = \sum_{\xi \in \mathcal{X}_a} w_a^0(\xi)\, \mathbf{1}[v \in \xi], \qquad q_v^L = 1 - \prod_{a \in L} \big(1 - q_v^a\big). \tag{11}$$

Every candidate extent contains its anchor, so $q_v^a = 1$ on a; it decreases outward as fewer and less probable candidates include v, and it is 0 outside the union of the candidates. The lesion marginal q_v^L combines the anchors of L as if their extents were independent. It summarises all candidate extents with their weights, so stage 2 does not commit to the single chosen extent. That extent is still defined for every anchor of an accepted or uncertain lesion, $\hat{\xi}_a = \arg\max_\xi w_a^0(\xi)$, and $\hat{\xi}_L = \bigcup_{a \in L} \hat{\xi}_a$ is the reference against which trimming and recall are reported.

Region of interest and lesion clusters. The model is defined on a region combining a local neighbourhood of the lesion with all of its credible hosts, because the host is uncertain and a tumor may extend into or arise from any of them:

$$R_L = \Big(\bigcup\nolimits_{a \in L} a\Big) \oplus r \;\cup\; \bigcup\nolimits_{h \in \mathcal{S}_L} \tilde{\Theta}_h, \tag{12}$$

restricted to the body. $\tilde{\Theta}_h$ is the territory of h, joined with its knowledge-base region when h is unsegmented or its mask is below the knowledge-base volume range (an organ largely replaced by tumor). The radius r is the 95th percentile, over expert tumors, of the distance from a simulated partial core to the farthest tumor voxel. If the posterior tumor reaches the boundary of R_L inside the body, R_L is enlarged by r around it and inference is repeated, so that a tumor much larger than its anchor is not cut by the region.

Stage 1 assigns each anchor to exactly one lesion, but the extents and regions of different lesions can overlap: one anchor's candidate extents may contain another anchor, sometimes of another lesion. Lesions whose regions overlap are grouped into a cluster $\mathcal{G}$ and inferred jointly on $R_\mathcal{G} = \bigcup_{L \in \mathcal{G}} R_L$. Clustered lesions may be parts of one heterogeneous tumor or neighbouring tumors; joint inference lets them share a boundary and appearance evidence instead of claiming the same voxels twice. The labels below are defined per cluster. Each lesion keeps its own hosts and its own spatial prior π_v^L (Eq. (13)); the cluster prior is $\pi_v^\mathcal{G} = 1 - \prod_{L \in \mathcal{G}} (1 - \pi_v^L)$, likewise $q_v^\mathcal{G} = 1 - \prod_{L \in \mathcal{G}} (1 - q_v^L)$, and the background classes of the cluster are the union of the lesions' classes $\mathcal{C}_L$, with their weights renormalised over the cluster. For a single lesion, $\mathcal{G} = \{L\}$ and $\pi_v^\mathcal{G} = \pi_v^L$.

Generative model. Let y_v be the image features of voxel $v \in R_L$: the intensity, smoothed in the acquisition plane, and local means and standard deviations at two scales. Each voxel carries a latent label $Z_v \in \{0, 1, \ldots, K_{\max}\}$: 0 is background, $1, \ldots, K_{\max}$ are tumor components. The tumor indicator is $t_v = \mathbf{1}[Z_v \geq 1]$, and the tumor mask is $\{v : t_v = 1\}$.

Spatial tumor prior. Tumors are not spatially homogeneous: they arise in particular organs, extend beyond the agents' outlines by characteristic distances, invade neighbours and respect some boundaries. A homogeneous label prior, as used for organ segmentation, does not express this. The prior probability that a voxel is tumor is

$$\pi_v^L = P(t_v = 1) = \sum_{h \in \mathcal{S}_L} P(h \mid L)\, \sigma\big(\eta^\top \psi_v(h)\big), \tag{13}$$

a mixture over the credible hosts. The features $\psi_v(h)$ are:

- the stage-1 extent marginal q_v^L (Eq. (11));
- the signed distance of v to the chosen extent $\hat{\xi}_L$;
- whether v lies in the territory of h, in another credible host, or in a confidently segmented organ outside $\mathcal{S}_L$;
- the depth of v below the surface of that organ.

The weights η are fitted on expert data only: expert tumors with simulated agent outlines (partial cores, dilated and leaking outlines, outlines merged with the organ, lost hosts) and expert organ masks. Every simulated outline is drawn around a real expert tumor, so Eq. (13) is the frequency with which an expert tumor occupies a voxel in that position relative to the agents' outlines and to the anatomy, *given that the lesion is a tumor*. It is a prior on extent, not on existence: no share of false outlines is simulated, because such a share would be a hand-set existence prior. The anchors enter only through q_v^L, which equals 1 on them. Because simulated anchors leak into normal tissue, the learned map gives them a prior probability below 1, and the image can still reject anchor voxels. Whether the lesion is a tumor at all is assessed by the control below. Unlike the extent prior of stage 1, Eq. (13) gives positive probability beyond the agents' outlines, which is what allows recall. That probability is learned near simulated outlines; far from any outline it is supported only by expert scans with several tumors, so new candidates far from the anchors are reported separately (below).

Adaptive spatial coupling. Neighbouring tumor indicators are coupled by a Potts term with latent pairwise weights, following the latent edge weights of BayeSeg [Gao et al., MedIA 2023]:

$$p(Z \mid \omega, \boldsymbol{\rho}) \;\propto\; \prod_v \big(\pi_v^{\mathcal{G}} \rho_{Z_v}\big)^{t_v} (1-\pi_v^{\mathcal{G}})^{1-t_v} \prod_{v\sim w} \exp\big(-\omega_{vw}\, s_{vw}\, \mathbf{1}[t_v \neq t_w]\big), \qquad \omega_{vw} \sim \mathrm{Gamma}(a_\omega, b_\omega), \tag{14}$$

where $\boldsymbol{\rho}$ are the component weights (below) and s_{vw} scales a neighbour pair by its physical distance, so thick slices couple less. The coupling acts on the tumor indicator only; components within the tumor are not spatially coupled. The image does not enter Eq. (14) directly. The weight ω_{vw} is inferred from the expected label disagreement of the pair, $q(t_v \neq t_w)$, which the appearance likelihood drives; where the image separates the two voxels, that disagreement is high and the posterior weight small. The boundary is therefore free to follow image contrast instead of a fixed smoothness. The hyperparameters (a_ω, b_ω) are estimated on expert tumor masks, from the frequency of label changes between neighbouring voxels.

Appearance and adaptive components. Tumor appearance and background appearance are latent and specific to the scan. Tumors are heterogeneous and long-tailed, so each component has a Student-t likelihood:

$$p(y_v \mid Z_v = k) = \mathrm{St}\big(y_v \mid \mu_k, \Lambda_k, \nu\big),\ k = 1..K_{\max}; \qquad p(y_v \mid Z_v = 0) = \sum_{c\in\mathcal{C}_{\mathcal{G}}} \lambda_{v,c}\, \mathrm{St}\big(y_v \mid \mu_c, \Lambda_c, \nu\big). \tag{15}$$

The background is a mixture over the credible hosts' normal tissue and the surrounding anatomy, $c \in \mathcal{C}_L$, pooled over a cluster into $\mathcal{C}_{\mathcal{G}}$. Its location weights $\lambda_{v,c}$ come from the organ territories, and the weight of host h is scaled by $P(h \mid L)$. The uncertainty about the host therefore carries into the voxel posterior instead of being resolved first. Each Student-t density is written as a Gaussian scale mixture, $y_v \mid Z_v = k, u_v \sim \mathcal{N}(\mu_k, (u_v\Lambda_k)^{-1})$ with a latent scale $u_v \sim \mathrm{Gamma}(\nu/2, \nu/2)$, which keeps the variational updates in closed form. The features are a vector, so the means and precision matrices (μ_k, Λ_k) have conjugate Normal–Wishart priors. Their hyperparameters are estimated on expert data, per organ class and modality: the contrast of expert tumors relative to their organ's normal tissue and its spread, and the variability of normal tissue.

No substructure annotations exist, so the number of components is not fixed. Given $t_v = 1$, the component follows weights $\boldsymbol{\rho} \sim \mathrm{Dir}(\alpha_0, \ldots, \alpha_0)$ with $\alpha_0 < 1$, a sparse prior under which unneeded components receive negligible weight and the effective number is inferred per lesion. The bound $K_{\max} = 4$ is the number of tissue types a radiologist distinguishes in a solid tumor: enhancing solid tissue, necrosis, cystic or fluid content, and calcification or haemorrhage. Components are unlabelled during inference; a name is attached afterwards by their appearance relative to the host's normal tissue and reported as a description, not used in any decision. On the scan, the appearance is initialised from the voxels weighted by $q_v^{\mathcal{G}}$ and inferred jointly with the mask.

Inference. The posterior $p(Z, u, \boldsymbol{\rho}, \theta, \omega \mid y)$, with θ the appearance parameters, is approximated by mean-field variational Bayes, $q(Z)\, q(u)\, q(\boldsymbol{\rho})\, q(\theta)\, q(\omega)$, with closed-form coordinate updates of the conjugate families [Gao et al., MedIA 2023]. The label posterior is initialised with $q(t_v = 1) = q_v^{\mathcal{G}}$, the cluster combination of q_v^L, so no single extent is chosen. The outputs are:

- the *voxel tumor posterior* $p_v = q(t_v = 1)$;
- the *component posteriors* $q(Z_v = k)$, an intra-tumoral substructure map, with the effective number of components;
- the *refined tumor* $\hat{Z}_L$, the connected components of $\{v : p_v \geq \frac{1}{2}\}$ that meet an anchor of L.

Comparing $\hat{Z}_L$ with the chosen extent gives the *trimmed* voxels $\hat{\xi}_L \setminus \hat{Z}_L$ (false positives of the agents' outline) and the *recalled* voxels $\hat{Z}_L \setminus \hat{\xi}_L$ (tumor the agents missed), each with the host territories it lies in. Components of $\{p_v \geq \frac{1}{2}\}$ that meet no anchor are reported separately as new candidates in the credible hosts, with their distance to the nearest anchor and without a decision; they can include benign findings such as cysts or vessels, and a rule for them is left until they have been reviewed. Lesions of a cluster whose refined tumors meet are reported as merged. Recall comes from the image and the spatial prior only: the agents' discarded proposals (claims, crumbs and lobes that stage 1 did not keep) are never re-admitted.

Control. Because no voxel is fixed, the model can explain the lesion's surroundings by background alone. The check is an approximate Bayes factor on the anchor neighbourhood $N_L = \big(\bigcup_{a \in L} a\big) \oplus r$, so that abnormal tissue elsewhere in a large host does not count for the lesion:

$$\widehat{\mathrm{BF}}_L = \exp\Big(\mathcal{L}_{N_L}(\text{tumor and background}) - \mathcal{L}_{N_L}(\text{background only})\Big), \tag{16}$$

where $\mathcal{L}$ is the variational lower bound on the log evidence and the background-only model is the same model with $t_v = 0$ everywhere. A ratio of lower bounds approximates the Bayes factor; it is not exact. If $\widehat{\mathrm{BF}}_L < 1$, the image prefers background alone: the lesion is flagged and its stage-1 extent is kept as the reported segmentation. The threshold 1 is where the two models are equally supported, and no other cut-off is used. The check does not change the stage-1 decision; it is reported with it.

Effect on the stage-1 evidence. Decisions are left unchanged. For every lesion, the stage-1 existence evidence (Section 2.3) is recomputed on $\hat{Z}_L$ under each host of $\mathcal{S}_L$, with the same expert tables and level α, and reported next to the stage-1 outcome, together with $\widehat{\mathrm{BF}}_L$ and $\bar{p}$ over the chosen extent and over $\hat{Z}_L$. Whether the refined evidence should feed back into decisions is decided after this comparison.

Estimation and validation on expert data. Every prior and hyperparameter of stage 2 comes from expert annotations, never from agent outputs or the evaluated cohort:

- the spatial prior η;
- the edge hyperparameters (a_ω, b_ω);
- the appearance hyperparameters, per organ class and modality;
- the degrees of freedom ν and the radius r.

$K_{\max}$ and α_0 are design values, disclosed; the sensitivity to α_0 is reported on R2Seg only. The data are R2Seg images with expert tumor masks, organ masks from TotalSegmentator and R2Seg, and simulated agent outlines. Validation holds out R2Seg collections in turn. It reports, per organ class and modality, voxel precision and recall, Dice, the over- and under-segmented fractions and the calibration of p_v, against the chosen extent, the marginal q_v^L thresholded at $\frac{1}{2}$, the anchor alone and the union of the agents' outlines. Stage 2 replaces the stage-1 extent only for the organ classes and modalities in which it improves the held-out Dice; elsewhere the stage-1 extent is kept, and this is reported.

Components. Without substructure annotations, components are evaluated against radiological reads. The TCGA-KIRC radiogenomic table (radiologists' reads distributed with the cohort; source citation to be added) records, per patient, solid composition, necrosis, calcification, margin and growth pattern; 95 of its patients have a segmented baseline series in our cohort (74 CT, 21 MR). The presence and volume fraction of each inferred component are compared with these reads. The reads are per patient, and some patients have several tumors, so lesions are matched to reads by side and tumor count, and patients whose tumors cannot be matched are left out and counted. The reads are used for evaluation only, never for fitting or choosing any setting, and no clinical or demographic variable of the table is used.